AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION

APRIL/ MAY 2019

Chemistry Paper II

Time: 2 hours 15 minutes Marks: 35

April/ May 2019
Teacher for Learning only

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's Signature**

RUBRIC

2. There are EIGHT questions. Answer ALL questions. Questions 7 & 8 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a simple calculator if you wish.

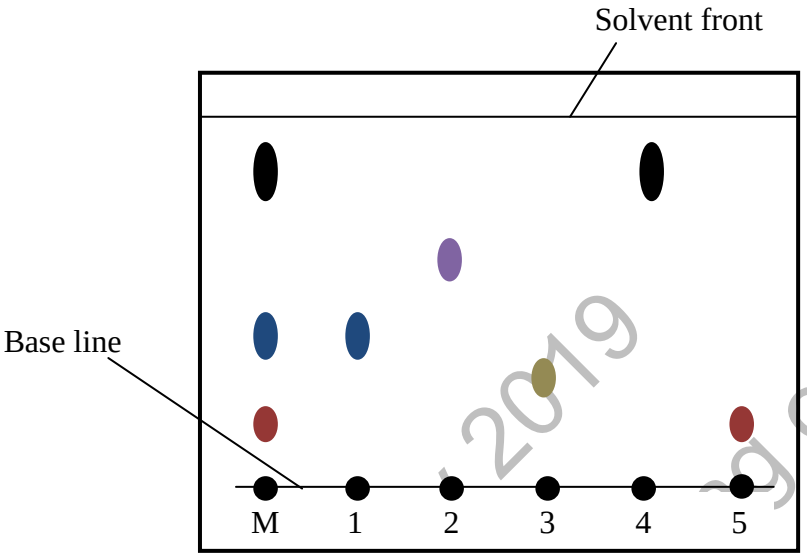
Q.1. (Total 4 Marks)

State any FOUR characteristics of an ionic compound.

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Q.2. (Total 4 Marks)

An analytical chemist uses chromatography to investigate the types of amino acids present in a mixture of amino acids. He represents the mixture as M, and the known amino acids as 1, 2, 3, 4 and 5. During experimentation, he ensures that the base line containing spots is placed above the solvent level. After experimentation, he sprays the chromatogram with a solution of ninhydrin. The given chromatogram shows his investigation result.



Chromatogram

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a. Why is the base line containing spots placed above the solvent level? (1 Mark)

b. What is the purpose of spraying ninhydrin solution on chromatogram? (1 Mark)

c. Based on his investigation, identify the amino acids which are present in the mixture. (2 Marks)

Q.3. (Total 4 Marks)

Write a reason for each of the following periodic trends observed in the periodic table.

a. Electronegativity increases across the period. (1 Mark)

_____

_____

b. Shielding effect increases down the group. (1 Mark)

_____

_____

c. Atomic radius decreases across the period. (1 Mark)

_____

_____

d. Ionisation energy decreases down the group. (1 Mark)

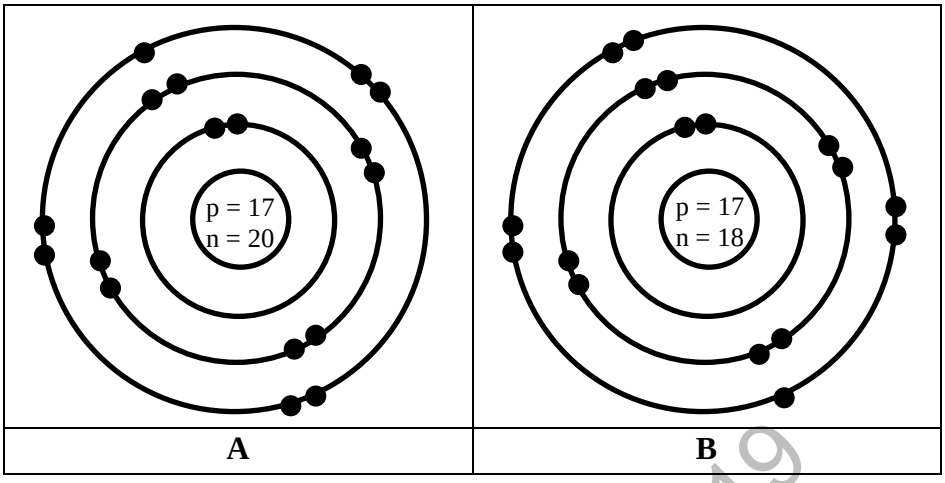
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Q.4. (Total 3 Marks)

Write any ONE similarity and TWO differences between the given atoms **A** and **B**.



Similarity between atoms **A** and **B**: _____

_____

_____

Differences between atoms **A** and **B**: _____

_____

_____

_____

_____

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Q.5. (Total 3 Marks)

Consider a closed container with liquid having weak intermolecular forces. This liquid experiences both, vapour pressure and boiling point at constant temperature.

a. Describe the effect of presence of weak intermolecular forces on the vapour pressure of the liquid. (2 Marks)

_____

_____

_____

_____

b. Relate the vapour pressure (described in part a) with the boiling point of the liquid. (1 Mark)

_____

_____

Q.6. (Total 3 Marks)

Give a scientific reason why gold is

a. used in making jewellery. (1 Mark)

_____

_____

b. alloyed with copper when making jewellery. (1 Mark)

_____

_____

c. extensively used in dentistry. (1 Mark)

_____

_____

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Q.7. (Total 7 Marks)

EITHER

a. A compound contains 24% carbon, 12% hydrogen and 64% oxygen by mass. The relative molecular mass of this compound is 50.057 g/mol.

i. Calculate the empirical formula of the given compound. (5 Marks)

ii. Determine its molecular formula. (2 Marks)

(**Note**: Atomic mass of C = 12 amu, H = 1 amu and O = 16 amu)

OR

b. Given below are four elements.

Non-Metal	Metal
Chlorine ($^{35}_{17}$Cl)	Magnesium ($^{24}_{12}$Mg)
Oxygen ($^{16}_{8}$O)	Potassium ($^{39}_{19}$K)

i. Identify TWO elements from the given table which, when form ions, show the same electronic arrangement as a neon ($^{20}_{10}$Ne) atom. (2 Marks)

ii. Illustrate the formation of an ionic compound using the elements identified in part i with the help of dot and cross structure of the compound. (5 Marks)

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Space for diagram for Q.7.b.ii.

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Q.8. (Total 7 Marks)

EITHER

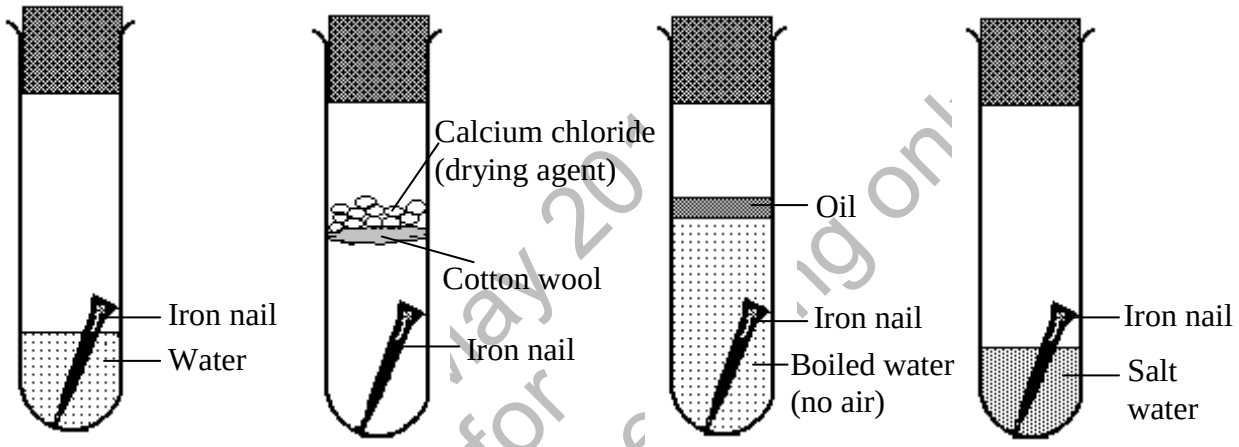
a.

i. Describe the arrangement and movement of particles in solids, liquids and gases. (6 Marks)

ii. What happens to the particles during effusion in gases? (1 Mark)

OR

b. The given diagram shows different conditions to investigate rusting of iron.



i. In which of the given test tubes, **I** to **IV**, will the iron nail rust? Explain why the iron nail rusts in the identified test tube(s) and not in others. (6 Marks)

ii. Why is the process of rusting a redox reaction? (1 Mark)

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END OF PAPER

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